CREDIT RISK ANALYSIS

(using MS-Excel)

Portfolio Project

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Project Overview

This project involves a credit risk dataset to demonstrate customer's demographics, financial status, employment and credit history analysis using Excel. It includes data cleaning, advanced formulas, pivot tables, lookup functions and dashboards for visualizing the results and identify key insights for making informative decisions.

Business Questions

In this project, several business questions are addressed using ad hoc analysis and Exploratory Data Analysis (EDA). These questions relate to customer demographics, loans, credit history, financial status, and employment history, as outlined below:

1. Which age group of customers contributes the most to credit risk?
2. What is the distribution of customers by marital status?
3. How many foreign workers fall under bad risk categories?
4. What is the distribution of male and female customers across different age ranges?
5. What is the distribution of credit risk across different checking account balances?
6. What is the distribution of credit risk across different saving account balances?
7. How many customers with different assets fall into the high credit risk category?
8. What is the distribution of credit risk among different housing status categories?
9. What is the distribution of bad credit risk across different types of credit history?
10. What is the distribution of bad credit risk across employment categories?
11. What is the distribution of customers across different employment year categories?
12. How is credit risk distribution across different loan duration categories?
13. How does credit risk vary by loan purpose?
14. How many customers fall into each category of savings and checking account balances?

Analysis Approach

For detail analysis of this dataset, Exploratory Data Analysis (EDA) and ad-hoc analysis are used.

1. **Exploratory Data Analysis**
   Performed Exploratory Data Analysis (EDA) on customer demographics, banking, and credit information to identify patterns, distribution, and understand relationships.

2. **Ad-hoc Analysis**
   Conducted ad-hoc analysis to answer specific business questions regarding credit risk, employment status, and loan distribution, providing actionable insights for decision-making.

Data Structure

1. **Raw Data**
   It's an original and unmodified dataset which serves the purpose of data integrity. This dataset is further used for data cleaning, transformation and dashboard analysis.

2. **Data Cleaning & Transformation**
   This process involves organizing and transforming the dataset to ensure data accuracy for further analysis. It includes changing data types, handling missing values, creating new columns, and organizing columns into multiple categories to support dashboard analysis.

- ***Organizing data***
  The dataset is structured to remove unnecessary spaces, duplicates and inconsistencies. The columns are structured/divided into different categories like:

  - Customer Demographics (Age, Gender, Marital Status etc.)
  - Residence & Assets (Owned Assets, Property etc.)
  - Banking Information (Saving Account, Checking Account etc.)
  - Employment (Job Type, Employment Duration etc.)
  - Credit Information (Credit History, Existing Credits etc.)
  - Loan Details (Loan Amount, Loan Duration etc.)
  - Credit Risk (Target variable, labeled as Good or Bad)

- ***Binning of Numerical Variables***
  Numerical columns like "Loan Duration", and "Age" are grouped into defined ranges to convert into segments for further analysis.

- ***Changing Data types:***
  Data types of each column are specified such that calculation can easily be done using pivot tables. The data types are shown in "Column Dictionary" sheet.

- ***Use of Advanced Formulas:***
  There are several advanced formulas that are used in the data transformation process such as VLOOKUP/XLOOKUP, conditional formulas, and logical operators. A separate Lookup Table sheet is created to make the binning process easier. The formulas are applied to columns such as "Age_Range", "Residential_Status", "Check_Acct_Balance", "Saving_Acct_Balance", "Skill_Type", "Employment_Type", "Years of Employment", and etc.

3. **Analysis using Dashboard & Pivot Tables**
   The dataset is divided into three segments for detailed analysis. Each business question is answered in the specific segment.

- ***Customer Demographics***
  The purpose of this analysis is to evaluate customer demographic factors such as Age, Gender, Marital Status, Number of Dependents and Foreign Workers status. To support this analysis, six Key Performance Indicators were defined, those are:

  - Total Customers - 1000
  - Bad Credit Risk – 30%
  - Good Credit Risk – 70%
  - Male – 33.6%
  - Female – 66.43%
  - Highest Credit Risk Age Range – 19-29

To further explain these KPIs, dashboards and pivot tables are used to identify the relationships and visualize the distribution between demographic variables and credit risk.

- ***Financial Status***

  This analysis helps to evaluate the financial condition of customer based on their banking and personal financial factors such as saving account status, checking account status, owned assets, and housing statuses. There are 5 Key Performance Indicators used for that:

  - Checking Acct. Balance Highest Bad Risk % - 49.3% - Negative Balance
  - Saving Acct. Balance Highest Bad Risk % - 36% - Low Savings
  - Owned Assets with Highest bad Risk – 102 – Cars
  - Top Bad Credit Risk for Housing Status – 186 - Owned
  - Bad Credit Risk Distribution among Acct. Balances – 114 – Negative Balance | Low Savings

  These KPIs are further presented through dashboards to help identifying patterns and relationships such as Checking acct. balance vs credit risk %, customer asset contribution for bad credit risk etc. These are defined to answer business questions to identify and highlight the factors that contributes in high-risk segments.

- ***Employment & Loan Analysis***

  The purpose of this analysis is to examine how employment-related factors and loan details impact customer credit risk. Key variables such as job type, employment duration, loan amount, loan duration etc. are considered to understand customer reliability.

  To support this analysis, KPIs are defined that includes:

  - Loan Purpose with Highest Bad Credit Risk – 38% - New Car
  - Highest Avg. Bad Credit Risk by Loan Purpose - $8209.33 - Others
  - Min. Credit - $250
  - Max. Credit – $18,424
  - Avg. Credit - $3,271.26
  - Top Bad Credit Risk Count by Credit History – 169 – Existing Credits Paid
  - Avg. Loan Duration (in months) – 20.9

  These KPIs are presented through dashboards to better understand the relationship between employment stability and loan behaviors. The analysis helps in identifying patterns that indicate higher credit risk, particularly among customers with unstable employment or higher loan burdens.

4. Lookup Tables

Lookup tables are created to simplify the categorization/binning process. Variables such as Checking account balance, saving account balance, employment years, Credit History and Age are grouped into more meaningful categories. These categorized values are then mapped back to the main dataset using lookup functions, ensuring consistency and improving the clarity of the data for analysis.

5. **Column Dictionary**
   The column dictionary provides a detailed description of each variable in the dataset, helping to ensure clarity and consistency during analysis. It serves as a reference to understand the meaning, and data type of each variable.

Tools & Techniques

1. Microsoft Excel: Pivot tables, formulas (VLOOKUP/XLOOKUP, IF, logical operators), charts
2. Data Cleaning & Transformation: Removing duplicates, correcting inconsistencies, categorization using lookup tables
3. Exploratory Data Analysis (EDA) to explore patterns and distributions
4. Ad-hoc analysis to answer specific business questions

Key Insights

1. **Demographics**

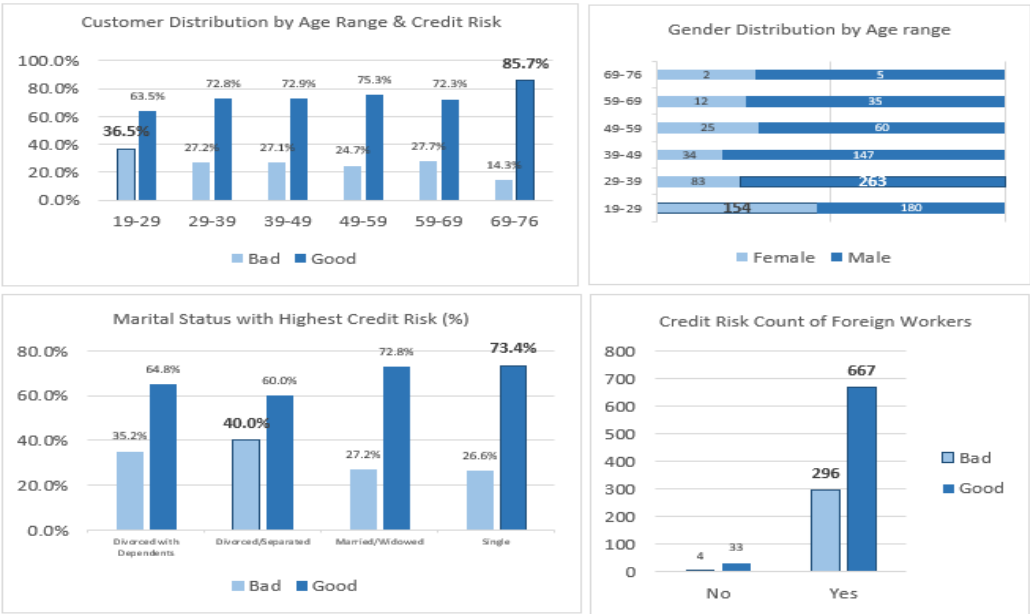

Figure 1: Dashboard 1 - Customer Demographics

- Customers aged 19-29 contributes to the highest bad credit risk of 36.5%.
- Customers who are divorced or separated contribute 40% to bad credit risk, indicating they are the top category among all marital statuses.
- A total of 296 customers classified as bad credit risk that lives in a foreign country while, only 5 customers live in the same country highlighting foreign workers have much highest credit risk.
- Males have the highest count of 263 in the 29-39 Age Range, while females top count is 154 in 19-29 indicating different age distribution between genders.

2. Financial Status

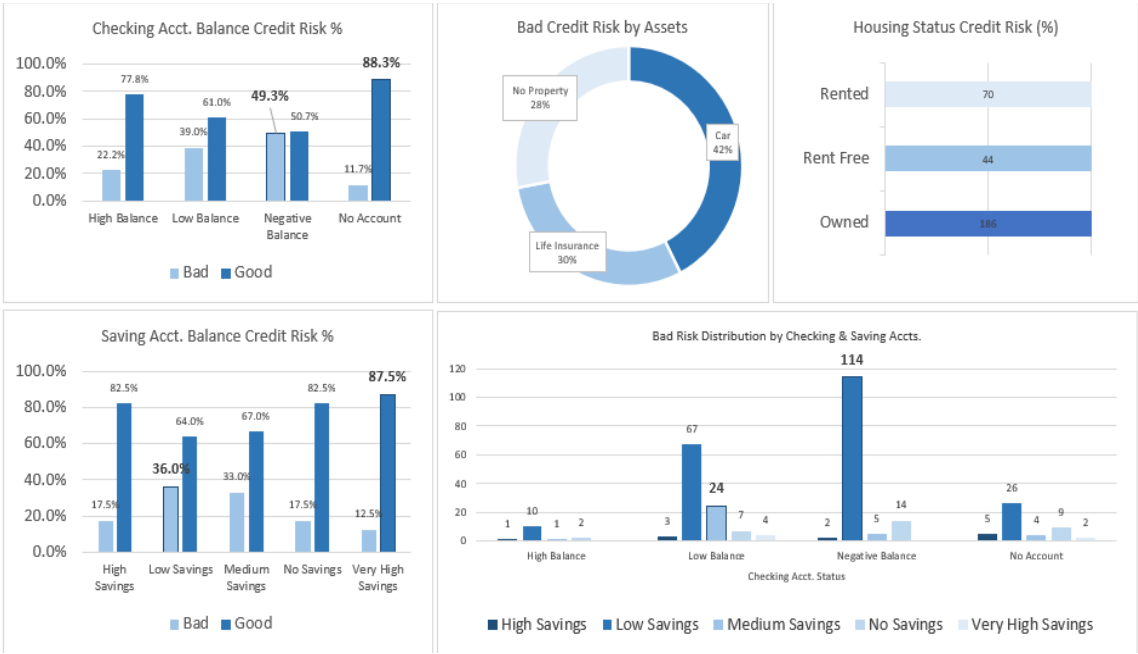

Figure 2: Dashboard 2 - Financial Status

- Customers with negative checking account balances contribute the highest proportion of bad credit risk at 49.3%, indicating that account overdrafts or negative balances are strongly associated with higher credit risk.
- Customers with Low saving account balances contribute the highest proportion of bad credit risk at 36%.
- Among owned categories, Cars have the highest bad credit risk at 42%, while Life Insurance and no property contribute up to 30% and 28%, respectively.
- Customers who own a house have the highest count at 186, while those in rented and rent-free categories have counts of 70 and 44, respectively, indicating that the majority of customers fall under home ownership.
- Customers with negative checking balances have the highest bad credit risk 114, while those with low balances have a count of 24.

3. Loan & Employment

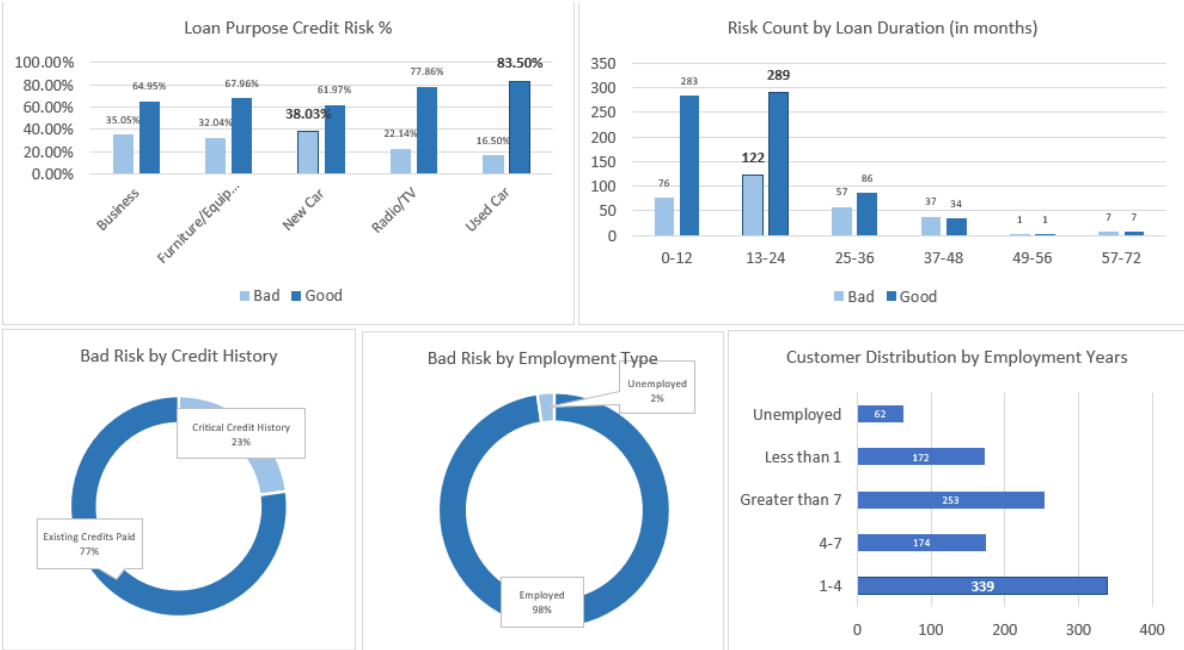

Figure 3: Dashboard 3 - Employment & Loan Analysis

- The highest number of customers (339) have 1–4 years of employment, indicating that most customers are relatively early in their careers.
- The highest number of bad credit risk customers (122) have loans with a duration of 13–24 months, indicating that medium-term loans are associated with a higher likelihood of default.
- Loans taken for a new car have the highest bad credit risk at 38.03%, indicating that this loan purpose is more likely to result in default compared to other purposes.
- Bad credit risk is highest among customers who have fully repaid existing credits 77% and those who are employed (98%), highlighting that credit history and employment type strongly influence risk levels.

Conclusion

This project focuses on credit risk analysis using ad hoc analysis and Exploratory Data Analysis (EDA) in Microsoft Excel. The goal is to explore customer demographics, financial status, employment history, and credit behaviour to uncover meaningful patterns and trends. Advanced Excel tools such as formulas, pivot tables, lookup functions, and charts, are used to organize and visualize the data effectively. The insights gained help decision-makers identify high-risk customers, reduce potential losses, and improve returns. Overall, this analysis demonstrates how data-driven approaches can guide smarter, more confident lending and risk management decisions.